

INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 1: Introduction and Research Problem

by

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1 Introduction and Research Problem

Decentralized Finance (DeFi) has transformed how capital is allocated on public blockchains: lending, borrowing, and leveraged trading can now run without traditional intermediaries ^[1]. Most credit-like transactions are still heavily collateralized, though, because protocols cannot draw on off-chain identity frameworks or existing credit data ^[2]. Under pseudonymity, the central issue is how to allocate capital efficiently among unknown parties.

This thesis asks what ERC-20 allowance edges add to PageRank as a measure of on-chain wallet reputation. The allowance edges come from the `Approval` events that approve and permit emit, and `EndorseRank` is PageRank computed on them with the edge weights of the transfer-based Adaptive Weighted PageRank (AWP) of Do, Do and Nguyen^[3]. Both are run on the same Arbitrum One wallets, together with a coupled walk over both edge types and the raw degrees that the walks smooth. Validation follows the rank-correlation method used with AWP^[3], with paired contrasts, raw-degree baselines and temporal holdouts added. In this thesis, an on-chain reputation metric is a value derived from on-chain activity and interactions without any off-chain identity information ^[4].

1.1 Background: DeFi, collateralization, and pseudonymity

Public blockchains make programmable settlement possible without a central authority ^[5]. A smart contract can encode the terms of a deal ^[6], and a DeFi platform that uses such contracts clears trades on the blockchain instead of routing them through a custodian ^[7]. Protocols then stack these contracts into decentralized markets for lending, automated market making and perpetual futures ^[8]. On Arbitrum One and other L2 networks, fees are low enough that ordinary users transact often, and every transfer, approval and protocol event can be inspected wallet by wallet.

Even with this visibility, DeFi credit markets remain inherently over-collateralized. Most protocols make a borrower or trader post more collateral than the position is worth, because the counterparty is a pseudonymous account whose owner has not been verified ^{[9],[10]}. Pseudonymity is a basic property of public ledgers. Anyone can create addresses at almost no cost, and one entity can own more than one wallet ^[11]. Conventional credit ratings depend on personal histories that translate poorly to this setting ^[2], so protocols control risk through collateral reserves, liquidation mechanisms, and protocol rules instead of personalized reputation.

Collateralization protects protocols, but it uses capital inefficiently. Traders and borrowers have to tie up funds they could otherwise invest, and the platforms offer none of the under-collateralized products that traditional finance extends to people with a credit record. Researchers have therefore turned to on-chain signals as a substitute for off-chain credit histories: transfer-graph centrality ^[4], network risk propagation ^[12], prospect theory and blockchain-based credit scoring ^[13], and machine learning pipelines over lending protocols ^[14]. Each of these lines of work assumes that public ledger data contains information usable for risk assessment without deanonymizing users.

Graph ranking algorithms appear well suited to this problem. PageRank and its variants rank nodes by recursive endorsement, so that a node's score depends on the endorsements it receives from other nodes ^[15]. Earlier web trust algorithms such as EigenTrust ^[16] and TrustRank ^[17] apply similar ideas to peer-to-peer trust and to web spam. Do and Do^[4] use both PageRank and HodgeRank on a graph of transfers to judge how trustworthy Ethereum users are, and call the result a form of social credit. Adaptive Weighted PageRank (AWP), proposed by Do, Do, and Nguyen^[3], is another transfer-graph ranking algorithm. It weights each transfer by a bounded function of its value and a logistic function of its age, and it restarts the walk at addresses that sent recently; its authors validate it by showing that it correlates with inbound transfer count and value, taken as proxies for economic importance. These studies establish transfer-graph reputation as a credible baseline, but they leave out explicit spending authorization, a separate on-chain action recorded as ERC-20 allowances ^{[18],[19]}.

1.2 Motivation for on-chain reputation

Four main factors drive the need for on-chain reputation research in leveraged DeFi.

- Capital efficiency: protocols are seeking ordinal ranking signals that could support tiered services (more frequent limit increases, lower fees, or higher priority in monitoring, for example), and perhaps even differentiated collateral requirements in future controlled experiments, while keeping settlement on chain ^[1].
- Auditability: off-chain credit scores are opaque, but graph-based scores computed from public event data can be recomputed and audited, which would help to ease some of the concerns about black-box decentralized credit scoring ^[2].
- Operational cost: a score that is refreshed often, before each leverage adjustment for instance, should be cheap to recompute.
- Standardization: other markets use screening ratios to tell counterparties apart, but pseudonymous DeFi has no standard ordinal language for doing so at the wallet level. Chapter 5 discusses where the scores studied here could serve that role and where they cannot.

Transfer-based AWP is public and auditable, but its graph grows with every payment ^[3]. Allowance graphs are a structurally different alternative. Whenever a user approves an ERC-20 token, the contract records how much of that token a spender may transfer ^{[18],[19]}, and each new approval supersedes the older ones, so the latest approval of each owner–spender pair and token stands for the history of that pair. Approvals are also rarer than transfers. On the same wallets the endorsement graph G_{approve} therefore has an order of magnitude fewer edges than the transfer graph G_{transfer} , and the PageRank solve, whose cost grows linearly with $|E|$, finishes proportionally faster (Table 4.1). The same solver runs in both cases, so the speedup comes from the sparser graph and is not an algorithmic improvement; Claim C1 treats it as an operational property of the representation. Efficiency aside, allowances also mean something different. A transfer might reflect routing, market making, or a passive receipt, whereas an allowance is an active grant of permission to spend

funds. Treating allowances as endorsement edges carries PageRank’s citation reading ^[15] over to a trust relationship that protocols already depend on. Whether this produces different rankings, and whether those rankings agree differently with authorization and transfer checks, is examined on a matched sample of $n = 5,521$ GMX V2 trading wallets on Arbitrum One.¹

1.3 Research problem

Adaptive Weighted PageRank (AWP) methods infer trust from transfer history through weighted sums and logistic time-decay ^[3]. Their economic interpretation is rich, yet AWP (i) must aggregate the whole transfer history, (ii) models a high-value wallet as one that receives many inbound transfers, and (iii) assigns no role to spending authorization. Our proposal, EndorseRank, keeps AWP’s edge weights and solver but builds the directed graph from the latest allowance of each owner–spender pair and token. The resulting graph is much sparser, and the signal it encodes is different. It also ranks a different side of each relationship. An allowance points from the owner to the spender, so EndorseRank scores the spenders that receive approvals, such as routers and vaults, and gives every owner that receives none the same value.

The main research question is: what do ERC-20 allowance edges add to PageRank as a reputation score for Arbitrum One wallets, compared with the transfer walk and with the raw degrees that both walks smooth? Research questions RQ1–RQ5 (Section 1.6) break it down. Because both edge types live on the same addresses, a single walk can also mix them node by node with a fixed weight λ , which defines a Coupled PageRank (C-PR); at $\lambda = 1$ and $\lambda = 0$ it walks one layer alone. The mixture is worth testing because of the unit of analysis: most traders are owners that an allowance score cannot tell apart, and one walk over both layers could keep the transfer layer’s coverage of them while carrying the allowance signal for spenders.

The scores appear in Table 1.1. Table 1.2 lists hypotheses (H1–H3), research questions (RQ1–RQ5),

¹Matched means that every method scores the same wallets: the GMX V2 wallets with at least three closes between 1 December 2025 and 31 May 2026, a list fixed from GMX data before any approval or transfer was extracted (Section 3.2).

and empirical claims (C1–C5) from Chapter 4.

Table 1.1: The scores evaluated in this thesis (matched wallet cohort, $n = 5,521$).

Score	Input graph	Role
EndorseRank	Latest ERC-20 allowances (G_{approve}), weighted as in AWP	Allowance delegation; the object of study in this re-search
AWP	Transfers (G_{transfer})	Transfer flow; the published transfer-graph method of Do, Do and Nguyen ^[3] (Section 3.6.1)
C-PR	Both layers, raw amounts, mixed per node with weight λ	Coupled operator used to test whether the two edge types combine (RQ5); at $\lambda = 1$ and $\lambda = 0$ it walks one layer alone. Its ablation is S-PR, a transfer walk restarted from the allowance walk
Raw degrees	G_{approve} and G_{transfer} , counted per node	In-approve degree and transfer in-degree, the local counts that the PageRanks smooth: proxies in the same window and baselines out of window

We define the main terms here to prevent any misreading:

- *DeFi (Decentralized Finance)*: Financial services such as lending and borrowing, implemented as smart contracts on public blockchains and generally run without a central operator in charge of custody or settlement ^[1].
- *Smart contract*: Self-executing program code that runs on a blockchain and carries out an agreement without a trusted intermediary ^[6]. In this thesis, the allowance and transfer logs are emitted by smart contracts.
- *Wallet / Address*: A blockchain account that holds funds and interacts with smart contracts; it is the object whose reputation we want to rank ^[7]. Each wallet is identified by its address, and no off-chain identity is assumed or recovered.
- *On-chain reputation score*: A number computed from on-chain data that expresses a wallet’s relative position for financial activity under pseudonymity ^[4]. Scores are ordinal and meant for ranking; their absolute scale has no meaning for credit limits.
- *PageRank*: A graph ranking algorithm that spreads influence through a directed graph with a damping factor ^[15]. Adaptive Weighted PageRank (AWP) is PageRank on the transfer

graph, with edges weighted by the value and age of each transfer and restarts at recently active senders ^[3]. EndorseRank applies AWP’s edge weights to the latest-allowance graph and restarts uniformly (Section 3.6).

- *Coupled PageRank (C-PR)*: One random walk over both the allowance and the transfer layer, which at each wallet mixes the two rows with weights λ and $1 - \lambda$ over the layers that wallet occupies (Section 3.6.3). Endorsement-seeded PageRank (S-PR) keeps the transfer walk but restarts it from the allowance walk, as TrustRank ^[17] restarts from a seed set.
- *ERC-20 allowance / Approve / Permit*: Contract methods that record how many of the owner’s tokens a spender may spend on the owner’s behalf ^{[18],[19]}. Approve is an ordinary on-chain call; EIP-2612 permit makes the same state change through a signature. For each owner–spender pair and token, EndorseRank uses the latest non-zero allowance.
- *Endorsement graph*: A directed graph whose edges denote an explicit delegation of spending authority, as opposed to an actual transfer of assets ^[18]. An edge $u \rightarrow v$ means that wallet u has approved wallet or contract v to spend tokens belonging to u .
- *Domain alignment*: The degree to which a reputation ranking agrees with an observable proxy ranking, measured with Spearman’s ρ ^[20] and Kendall’s τ ^[21] and interpreted as construct validity in the sense of Cronbach and Meehl^[22]. A method can align closely with the graph of its own construct and weakly with a different check.

1.4 Temporal holdout

The contemporaneous allowance Kendall τ (Table 4.3) is an intended-construct check. PageRank on G_{approve} should agree with spender-side approval degree, because the score is a global smoothing of that degree. To test whether the score conveys information beyond the same-window graph, the scores are frozen at 28 February 2026 and compared with two labels that only become available in March–May 2026: new owner–spender approval pairs (the endorsement construct) and new transfer senders (the flow construct). The holdout cohort ($n = 1,335$) consists of every spender

that holds at least one positive latest allowance at t_1 (28 February 2026, 23:59:59 UTC) in the extracted approval data. Each PageRank is also compared with the raw degree it smooths at t_1 , so the gain of the propagated score over the local count can be seen directly.

A second window was designed on 28 September 2026, after the results of this spring holdout were known, and registered on 1 October 2026. Its scores are frozen on 31 May 2026 and its labels are counted from June to August 2026, months whose logs had not been extracted when the registration was committed. It repeats the two spender labels and adds one for the matched traders: the share of their GMX closes that did not end in liquidation (Section 3.7.2).

1.5 Research objectives

The research has five objectives, one for each of RQ1–RQ5. Each is stated with the measurement used to operationalize it and with the criterion that Chapter 5 applies when judging whether the objective was met.

- O1. Establish whether EndorseRank and AWP order the wallets of the same cohort differently (RQ1). Measurement: Kendall τ between the two score vectors ($n = 5,521$) with a 95% bootstrap interval. Criterion: the interval contains neither 0 (unrelated orderings) nor 1 (identical orderings).
- O2. Assess how closely each score agrees with each construct in the same window (RQ2). Measurement: family-level Kendall τ with 95% intervals for every score and construct. Criterion: a score agrees with a construct when its interval on that family excludes 0; agreement with the construct of the score’s own edges is read as an intended-construct check, because those proxies come from the same events as the graph.
- O3. Measure what each score costs to compute on the same wallets, and how runtime responds to the number of sampled wallets (RQ3). Measurement: solve time (mean and standard deviation of five timed runs after one warm-up run), memory, iterations, $|V|$ and $|E|$, all under one protocol. Criterion: the time ratio is reported only together with the ratio of $|E|$,

Table 1.2: Summary of hypotheses, research questions, and empirical claims.

Item	Addresses	Reported in
H1 / RQ1	Do EndorseRank and AWP produce different wallet orderings?	§4.4; inter-method τ with 95% CI; Table 4.5; Claim C3
H2 / RQ2	How closely does each score agree with the allowance checks and with the transfer checks?	Tables 4.3 and 4.5; Claim C2
H3 / RQ3	What does each score cost to compute at equal n ?	Tables 4.1–4.2; Claim C1
RQ4	Do t_1 scores predict t_2 new approvals and new senders, and does any PageRank do better than the raw degree it smooths?	§4.6; Tables 4.10, 4.11, 4.12, 4.13 and 4.16; Claim C4
RQ5	Does the coupled operator C-PR improve on a walk over the transfers alone, and does it rank wallets better than a walk over either layer alone?	§4.3.1, §4.6 and §4.6.2; Tables 4.3, 4.4, 4.11, 4.14 and 4.15
C1	Runtime and graph size of each score	Table 4.1
C2	Same-window agreement of each score with the checks of its own edge type, partly built in and, for EndorseRank, set by its ties	Tables 4.3 and 4.5
C3	EndorseRank and AWP order the cohort differently, mainly through which wallets are in the allowance graph	Table 4.5; Figure 4.3
C4	Out of window no PageRank ranks a label as well as the raw degree of the same edge type, in either window	Tables 4.10, 4.11, 4.12 and 4.16
C5	Coupling the two edge types does not beat its two layers walked alone, and the registered replication does not confirm its exploratory gain over the transfer walk: the gain on new approvals replicates, but non-inferiority on new senders fails, and against AWP it fails clearly	Tables 4.3, 4.4, 4.11 and 4.15; §4.6.2

so that any difference can be traced to the size of the graph.

- O4. Test whether scores frozen at t_1 predict new approvals and new transfer senders at t_2 among the cohort's spenders, and whether any PageRank does better than the raw degree it smooths (RQ4). Measurement: Kendall τ with 95% intervals between every score computed on 28 February 2026, the two raw degrees included, and both labels counted from March to May 2026 on the $n = 1,335$ spenders, with paired contrasts of each PageRank against its own degree. Criterion: a score predicts a label when its interval excludes 0, and a PageRank improves on its degree only when the interval of the paired difference lies above 0.
- O5. Test whether one walk across both edge types improves on a walk over the transfers alone, and whether it beats a walk over either layer alone (RQ5). Measurement: C-PR at $\lambda = 0.5$ (with 0.25 and 0.75 as sensitivity values) and S-PR, compared through paired intervals on $\Delta\tau$ with the two layers walked alone and with EndorseRank and AWP, on the holdout and same-window labels. Criteria, each fixed before the data it is judged on: (a) the rule of 14 September 2026, under which C-PR does no worse than the allowance walk on new approvals and better than the transfer walk on new senders on the spring holdout, and in the same window lies within the interval of the leading layer on the transfer and allowance families and above both layers on Sybil stability; (b) the rule registered on 1 October 2026, under which C-PR beats the transfer walk on new approval pairs in the June–August window and is not worse by more than 0.02 on new senders or on the traders' liquidation-free close rate. A registered sensitivity analysis repeats (b) with AWP in place of the transfer walk; it does not change the rule.

1.6 Research questions

Three hypotheses go with the first three research questions. H1: on the same wallet set, EndorseRank and AWP produce different wallet orderings. H2: each score agrees with the checks built from the edges it walks on. H3: at the same value of n , a score's computation time follows the number

of edges in its graph, so EndorseRank, on the sparsest graph, is the fastest to compute. RQ4 and RQ5 are answered by contrasts and rules fixed before the relevant data (Section 3.7). Table 1.2 summarizes the questions.

- RQ1. On the same Arbitrum wallet set, do EndorseRank and AWP produce different wallet orderings?
- RQ2. How closely does each score agree with current allowance checks and with current transfer checks (Spearman's ρ and Kendall's τ)?
- RQ3. What does each score cost to compute on the same wallet set, as measured by time, memory, and iterations?
- RQ4. Given scores computed at time t_1 , do the scores predict t_2 new owner–spender approvals to inbound spenders and new transfer senders, and does any PageRank predict them better than the raw degree it smooths at t_1 ?
- RQ5. Does a coupled PageRank running on both the allowance and the transfer layer improve on a walk over the transfers alone, such as AWP, and does it rank wallets better than a walk over either layer alone, out of sample as well as in sample?

RQ1 compares the orderings of the two single-layer scores. Under RQ2 every own-construct proxy is computed from the same events as the graph, so RQ2 is an observation about construct validity and not external validation. RQ3 measures cost on the same wallet sets and how it responds to node count. RQ4 keeps endorsements as the construct and tests the prediction of future new approvals, with new transfer senders as a second label; the raw degree at t_1 is the baseline that each PageRank has to beat. It replaces the trading-success validation of the research proposal, whose outcome-based reference was circular (Section 2.8). RQ5 moves from comparing the two edge types to combining them. Its rule for beating both layers was fixed before any coupled score was computed; its rule for improving on the transfer walk came after the spring labels had been seen, so it is judged only on the June–August window registered on 1 October 2026 (Sections 3.7 and 3.7.2).

1.7 Summary of the literature review

Chapter 2 reviews four areas. In graph ranking, PageRank^[15] ranks nodes by recursively propagated endorsement at a per-iteration cost linear in the number of edges^[23], and EigenTrust^[16] and TrustRank^[17] adapt it to peer-to-peer trust and web spam with normalization and seed sets. In on-chain reputation, Do and Do^[4] apply PageRank and HodgeRank to Ethereum transactions, and AWP^[3] weights transaction edges by value and age and validates its ranking against inbound transfer count and value, statistics drawn from the same data as its graph. Lin et al.^[12] propagate account risk through the transaction network, and surveys of blockchain reputation systems^{[25],[26]} find explicit ratings and transfer volume to be the usual trust signals. The tooling deployed in DeFi works differently again: Human Passport aggregates credentials about the holder of an address, on-chain and off-chain, to decide whether the address belongs to a distinct person^[64]. None of this work takes the ERC-20 allowance^[18] as a graph edge. In DeFi risk and measurement, surveys^{[8],[27]} place behavioral counterparty risk beside contract, oracle, market and governance risk, and wash trading^{[28],[29]}, extractable value^[30], scam tokens^[31] and airdrop farming^[32] show that much transfer activity is artificial. In measurement theory, construct validity^{[22],[33]} separates what a score measures from external criteria, and the bootstrap^{[34],[35]} gives distribution-free intervals for rank statistics. The gap (Section 2.21) is an allowance-edge ranking evaluated on one cohort beside the transfer method and the raw degrees, with confidence intervals, paired differences, a fixed timing protocol and a temporal holdout.

1.8 Summary of the research methodology

Chapter 3 applies a quantitative, comparative design to public ledger data. ERC-20 Approval and Transfer logs on Arbitrum One from 1 December 2025 to 31 May 2026 are extracted from Google BigQuery for a matched cohort of $n = 5,521$ GMX V2 wallets, fixed from GMX data alone. One power-iteration PageRank implementation scores the latest-allowance graph and the transfer graph, and the coupled operator runs on the same solver. Each score is compared with

Spearman’s ρ and Kendall’s τ against same-window transfer, allowance and Sybil-stability proxies and, out of window, against new approvals and new transfer senders on the spender cohort, with the raw degrees at the freeze as baselines. A registered second window repeats the holdout from June to August 2026 and adds a trader label. Every coefficient has a 95% interval from 400 paired wallet resamples (seed 42), and method comparisons are paired. Runtime, memory, iterations and graph size are recorded under one timing protocol. The code, configuration and machine-readable summaries behind every table are in a public repository (Section 3.13), and the study uses only public data (Section 3.16).

1.9 Contributions

The thesis makes four contributions to the study of on-chain reputation and DeFi risk. EndorseRank and AWP both apply weighted PageRank ^[15] to a single edge type, each with its own restart distribution; only the coupled operator changes the transition rule. What is new is therefore an edge type, an evaluation design, and a registered test of whether the two edge types combine.

- Latest-allowance edge construction (EndorseRank): an endorsement graph in which an edge $u \rightarrow v$ stands for the latest in-window ERC-20 allowance, set by approve or permit, that the spender v holds from the owner u , weighted as AWP weights a transfer. The edge records a delegation of spending rights, not a history of payments, and none of the reputation methods reviewed uses it (Section 2.21).
- Same-cohort evaluation with statistical inference: every score of the thesis is computed on one set of wallets and checked against identical transfer, allowance and Sybil-stability proxies under the AWP validation protocol ^[3], with 95% bootstrap intervals for every coefficient and paired intervals for the contrasts ^[34]. Part of each score’s agreement with its own proxies is built in, and runtime is reported beside edge counts because Claim C1 follows from sparsity.
- Temporal holdout with degree baselines: scores frozen at t_1 are tested against the new approvals and new transfer senders that arrive after t_1 , beside the raw degrees that PageRank

smooths. The design separates in-sample agreement from out-of-sample prediction, and the propagated score from its local count; with the open pipeline (Appendix 8.1) it provides the out-of-window evidence behind Claims C4 and C5.

- Coupled operator and its registered evaluation: C-PR blends a raw-amount allowance layer and a raw-amount transfer layer node by node with a fixed weight λ , and S-PR is a seeded ablation. A first rule, fixed with λ before any coupled score was computed, asked C-PR to beat both layers walked alone. A second rule, registered before the logs of its window were extracted, asked it to improve on the transfer walk. Both fail, and the thesis reports them as failed (Claim C5). The ablation adds one more result: the allowance signal survives when allowance edges carry the walk and is lost when they only decide where the walk restarts.

1.10 Scope and limitations

The study covers Arbitrum One from 2025-12-01 to 2026-05-31. Its main sample is the $n = 5,521$ GMX V2 wallets with at least three closes in that period, and its holdout sample the $n = 1,335$ spenders that held a positive allowance on 28 February 2026. EndorseRank scores a wallet by the approvals it receives, so it separates spenders and not the owners who grant approvals; most of the matched traders are owners, which is why the holdout is run on spenders, and the registered June–August window adds a trader label (Section 3.7.2). Runtime is also measured on subgraphs of the cohort graph drawn through a larger address pool ($N = 99,080$) that adds no edges. Default labels in unsecured lending are outside the scope of this study, extended baselines are archived (Appendix 8.4), and no claim is made that the results generalize beyond Arbitrum One or the months studied. Amounts are raw token base units. EndorseRank and AWP pass them through a bounded function, so every approval or transfer of ten base units or more counts about the same, while in the raw layers of C-PR unlimited approvals decide the rows of many owners (Chapter 3). Adversarial manipulation of allowances or transfers is a real threat: Chapter 5 shows that a closed farm of Sybil addresses can inflate EndorseRank cheaply, and the Sybil-adjusted proxies describe some

manipulation patterns without protecting the score against them ^[11].

1.11 Research roadmap

Chapter 2 reviews graph-based on-chain reputation, ERC-20 allowance semantics and the reputation tooling deployed in DeFi, and develops the theoretical framework. Chapter 3 defines the cohorts, the graphs and scores, the construct checks, the temporal holdouts, the timing protocol, and the standards for reproducibility and ethics. Chapter 4 reports the benchmark (Claim C1), the same-window alignment of every score (C2), rank divergence (C3), robustness checks and the temporal holdouts with the registered replication (C4 and C5); Appendix 8.5 collects the direct comparisons between EndorseRank and AWP. Chapter 5 interprets the findings by research question and objective, works out their implications, models the cost of Sybil attacks on allowance graphs, and weighs threats to validity and limitations. Chapter 6 answers RQ1–RQ5, states the contribution to knowledge and sets out future work.